TOMMASO CRIPPA

High Performance Computing Engineering Student

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HPC Engineering student with hands-on experience building data pipelines and ML-backed analytics tools. Proficient in Python, SQL, and Kafka, seeking to apply these skills in scalable data infrastructure and ETL processes.

EDUCATION

Double Master's Degree in High Performance Computing — **Politecnico di Milano - University of Luxembourg**
29.16 / 30 — 2024 - 2026

- Selected for the exclusive EUMaster4HPC double-degree programme, in which I participated in extracurricular HPC challenges, an international summer school and workshops with industry leaders

Bachelor degree in Computer Engineering — **Politecnico di Milano**
110 cum laude / 110 — 2021 - 2024

- Won "Migliori Matricole" scholarship for being one of Politecnico di Milano's best freshmen students of the 21/22 season.

Relevant courses completed: Streaming Data Analytics, Time Series Analysis, Artificial Neural Networks and Deep Learning

EXPERIENCES

Master's Thesis Intern @ MEGWARE Computer Vertrieb und Service GmbH — **Chemnitz**, Germany
HPC | Machine Learning | Time-Series | SLURM | Linux — February 2026 - August 2026

Crippius/hpc-bottleneck-detector

- Built an end-to-end weakly-supervised pipeline that automatically labels and classifies 100+ HPC metrics sampled at ∼5,000 data points per minute, with zero human annotation required.
- Validated against HPAS fault-injection ground truth, reaching 0.954 macro-F1 in cross-validation and generalizing well to unseen hardware with no retraining.

Operations & Logistic @ Google Developer Group On Campus - Polimi — **Milan**, Italy
AI | Cloud | Mobile — August 2024 - August 2025

- Organised and executed 5+ technical events over 12 months on AI, Cloud Computing, and Mobile Development, including a hackathon with 150+ participants, doubling community membership over the year.

PROJECTS

HPC Streaming Telemetry Pipeline
Python | Kafka | Spark | TimescaleDB | Docker | Grafana — Crippius/hpc-streaming-telemetry

- Built an end-to-end streaming pipeline replaying Marconi100 supercomputer telemetry (6.3M events) through Kafka and Avro into Spark Structured Streaming and TimescaleDB, sustaining 74k messages/s at 3.3 s median latency, with idempotent upserts producing no lost or duplicated windows after a mid-run Spark kill.

Benchmarking Detectors in Streaming Machine Learning
Python | capymoa | pandas | numpy | matplotlib | Colab — Streaming-Data-Analytics/Benchmarking-Detectors-in-SML

- Benchmarked concept drift detectors across 4 streaming datasets (∼3M data points total) covering abrupt, gradual, incremental, and recurrent drifts, evaluating accuracy, Kappa, and runtime to identify best-fit detectors per scenario.

Whatsapp Wrapped
Python | matplotlib | pandas | fpdf | Flask | HTML | CSS | Javascript | Vercel | Render — Crippius/whatsapp-wrapped

- Built a full-stack web app that transforms WhatsApp chat exports into personalised analytics reports (message trends, word frequency, response times), deployed on Vercel and Render with 100+ users since launch.

Early Alzheimer Detection with Deep Learning
Python | pytorch | keras | nilearn | C++ | libtorch | clinica | SLURM — justvicthor/Early_Alzheimer_Detection_DL

- Developed a deep learning model for early Alzheimer's detection, achieving Macro-AUC of 0.9197 on the test set, improving upon the original paper (0.80), with full data preprocessing and augmentation pipelines run on the MeluXina supercomputer.

SKILLS AND LANGUAGES

Tech Stack: Python (Advanced), Java, Kafka, SQL (Advanced), Spark
Infrastructure: Git (Advanced), DVC, Docker, AWS
Languages: Italian (Native), English (Proficient – IELTS 7.5), German (Beginner – A2)